Exercise 4A: Exponential families

Key exercises are marked with (*).

1. (*) Explain the following concepts:
   a) Exponential family b) Sufficient statistic c) Natural parameter
   d) Link function e) Score function

2. (*) The probability mass function of the Poisson distribution is

$$f(x;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

   a) Write down the log-likelihood of a set of N independent and identically distributed Poisson variables.

   b) Explain what "independent and identically distributed" (iid) means, and how the iid assumption shows up in this log-likelihood equation.

3. (*) Express the following distributions as exponential families:

   a) Normal distribution

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

   b) Binomial distribution

$$f(k;n,p) = \binom{n}{k} p^k (1-p)^{(n-k)}$$

4. In CASI, exponential families are expressed in this form:

$$f_\lambda(y) = \exp(\lambda y - \gamma(\lambda)) f_0(y) \tag{1}$$

   Another common formulation is as follows:

$$f(x|\theta) = h(x)g(\theta)\exp(\eta(\theta)\cdot T(x)) \tag{2}$$

   Show that the two are equivalent, so any distribution that can be expressed in form 1 can also be expressed in form 2. Which symbols in Eq. 1 correspond to which in Eq. 2?

5. In Figure 8.2 of CASI, the numbers of mice dying in the 11 groups were 0, 0, 0, 3, 6, 6, 5, 9, 9, 10, 10. Use the Python package `statsmodels` to calculate the red logistic regression curve and replicate the plot.